AS1109 Mid-Sem Examination 2026

Complete Student Solution Key

B.Tech., Semester I – Calculus and Ordinary Differential Equations

Max. Marks: 30 Time: 2 hours

Purpose of this key. Every question from the paper is reproduced below, followed by a slow, step-by-step solution. The coloured boxes tell you *what idea to use*, *what calculation to do*, and *what conclusion to write*. Use this later as a tutorial/revision document rather than memorising individual answers.

Read the question → Choose the idea → Calculate carefully → State the conclusion

Q1. Limits, partial derivatives, and differentiability [2 marks]

Q1(a) – Limit versus value at the point [1 marks]

If

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L,$$

must f be defined at (x_0, y_0)? Give reasons for your answer.

Idea

A limit asks what happens to $f(x,y)$ at points *near* (x_0, y_0). The value exactly at (x_0, y_0) is not used in the limit.

The statement

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$$

means that the values of $f(x,y)$ can be made arbitrarily close to L by taking (x,y) sufficiently close to (x_0, y_0), with $(x,y) \neq (x_0, y_0)$.

A simple example is

$$f(x,y) = x^2 + y^2, \qquad (x,y) \neq (0,0),$$

with $f(0,0)$ left undefined. Even though the function has no value at the origin,

$$\lim_{(x,y)\to(0,0)} (x^2 + y^2) = 0.$$

No. f need not be defined at (x_0, y_0). A limit depends on nearby values, not on the value at the point itself.

Q1(b) – Do two partial derivatives guarantee differentiability? [1 marks]

Consider a function $f(x,y)$ and a point (a,b). Suppose that the partial derivatives $f_x(a,b)$ and $f_y(a,b)$ both exist. Does this necessarily imply that $f(x,y)$ is differentiable at (a,b)? Give reasons for your

answer.

Idea

Differentiability is stronger than the existence of the two partial derivatives. The partial derivatives only inspect motion parallel to the coordinate axes.

Consider

$$f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Along the x-axis, $f(h,0) = 0$, so $f_x(0,0) = 0$. Along the y-axis, $f(0,k) = 0$, so $f_y(0,0) = 0$.

But along the line $y = x$,

$$f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2} \qquad (x \neq 0).$$

Hence $f(x,y)$ is not even continuous at $(0,0)$, so it certainly cannot be differentiable there.

No. The existence of $f_x(a,b)$ and $f_y(a,b)$ alone does not guarantee differentiability. Differentiability requires a genuine linear approximation in *all* nearby directions.

Do not write "partials exist, therefore differentiable." A useful sufficient condition is that the first partial derivatives exist in a neighbourhood and are continuous at the point.

Q2. Domain, boundary, openness, and range [3 marks]

Q2

Consider

$$f(x,y) = \frac{1}{\sqrt{x^2+y^2-36}}.$$

a) Determine and sketch the domain of f.

b) Find the boundary of the function's domain. State whether the domain is open or closed.

c) Find the range of f.

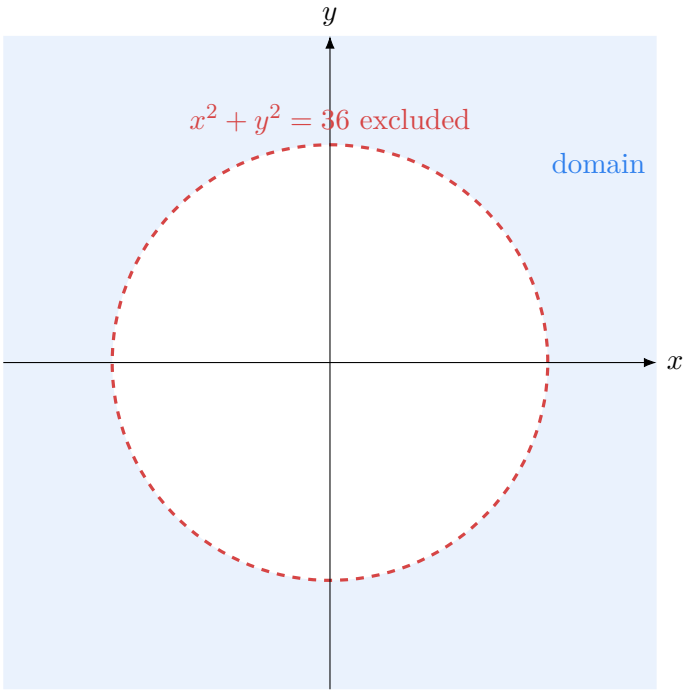
The square root must be real, and because it is in the denominator it must also be non-zero. Therefore

$$x^2 + y^2 - 36 > 0.$$

Equivalently,

$$\boxed{x^2 + y^2 > 36}.$$

So the domain is the exterior of the circle $x^2 + y^2 = 36$ (radius 6), with the circle itself excluded.



The boundary is exactly the circle

$$\boxed{x^2 + y^2 = 36}.$$

Every point in the domain satisfies the strict inequality $x^2 + y^2 > 36$, so each domain point has a small disk around it that stays inside the domain. Hence the domain is **open**.
It is not closed because it does not contain its boundary circle.

Let

$$t = \sqrt{x^2 + y^2 - 36}.$$

On the domain, t can be *any* positive real number. Therefore

$$f = \frac{1}{t}$$

can also be any positive real number.

$$\boxed{\text{Range}(f) = (0, \infty)}.$$

As $x^2 + y^2 \to 36^+$, $f \to \infty$. As $x^2 + y^2 \to \infty$, $f \to 0^+$, but f never actually equals 0.

Part	Credit	Marks
(a)	Correct condition $x^2 + y^2 > 36$ with reason	$\frac{1}{2}$
(a)	Correct sketch: exterior of radius-6 circle, boundary excluded	$\frac{1}{2}$
(b)	Boundary $x^2 + y^2 = 36$	$\frac{1}{2}$
(b)	Correctly states domain is open, with reason	$\frac{1}{2}$
(c)	Range $(0, \infty)$	1

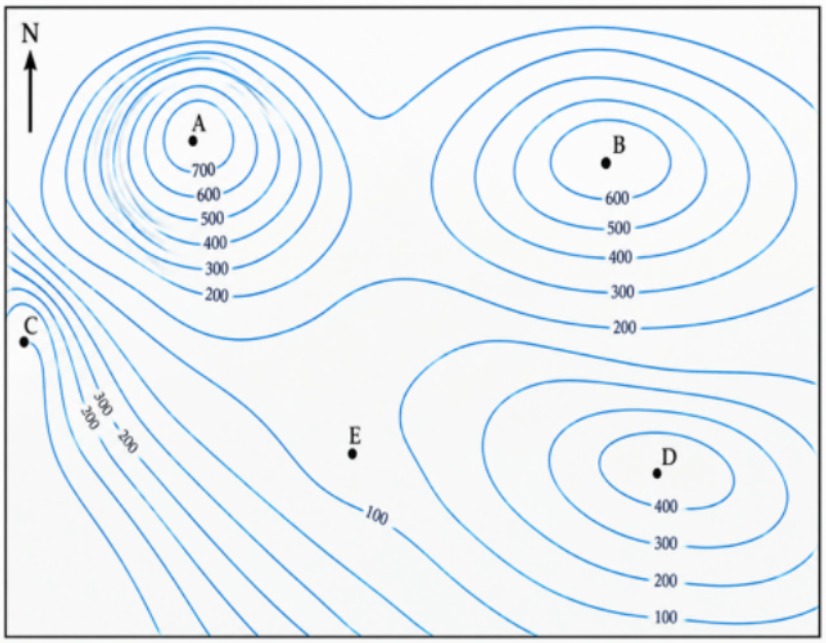
Q3. Reading a contour map [1 marks]

Q3

The figure shows a contour map of a region. The contour interval is 100 metres, and the numbers indicate elevations above sea level.

a) Determine which peak has the highest elevation.

b) At which location is the slope steepest?



Idea

On a contour map:

- higher contour numbers mean higher elevation;
- closer contour lines mean a larger change in elevation over a shorter horizontal distance, i.e. a steeper slope.

Around A, the contours climb through $200, 300, 400, 500, 600, 700$ m before reaching the peak. Around B, the highest labelled closed contour is lower. Therefore the highest peak is A.

The contour lines are packed most tightly near C. Therefore the elevation changes most rapidly with horizontal distance there.

Highest peak: A Steepest slope: C

Q4. Implicit partial differentiation [2 marks]

Q4

Find $\partial z/\partial x$ at the point $(1,1,1)$ if

$$xy + z^3 y - 2zxy = 0$$

defines z as a function of the two independent variables x and y and the partial derivative exists.

Idea

Differentiate with respect to x while treating y as a constant and remembering that $z = z(x,y)$.

Starting from

$$xy + z^3 y - 2zxy = 0,$$

we differentiate with respect to x:

$$y + 3z^2 y\, z_x - 2(zy + xy\, z_x) = 0.$$

At $(x,y,z) = (1,1,1)$,

$$1 + 3z_x - 2(1 + z_x) = 0.$$

Thus

$$1 + 3z_x - 2 - 2z_x = 0 \quad \Longrightarrow \quad z_x = 1.$$

$$\boxed{\left.\frac{\partial z}{\partial x}\right|_{(1,1,1)} = 1}$$

A very compact alternative is to set $F(x,y,z) = xy + z^3 y - 2zxy$. Then $z_x = -F_x/F_z$. At $(1,1,1)$, $F_x = -1$ and $F_z = 1$, hence $z_x = 1$.

Q5. Linear approximation in three variables [2 marks]

Q5

Find the linear approximation $L(x,y,z)$ to

$$f(x,y,z) = \ln(x + yz)$$

at the point $P(2,1,-1)$.

Idea

For a function of three variables,

$$L = f(P) + f_x(P)(x-a) + f_y(P)(y-b) + f_z(P)(z-c).$$

$$f(2,1,-1) = \ln(2 + 1 \cdot (-1)) = \ln 1 = 0.$$

$$f_x = \frac{1}{x+yz}, \qquad f_y = \frac{z}{x+yz}, \qquad f_z = \frac{y}{x+yz}.$$

At $P = (2,1,-1)$, the denominator is 1, so

$$f_x(P) = 1, \qquad f_y(P) = -1, \qquad f_z(P) = 1.$$

$$\begin{aligned} L(x,y,z) &= 0 + 1(x-2) - 1(y-1) + 1(z+1) \\ &= x - y + z. \end{aligned}$$

$$\boxed{L(x,y,z) = x - y + z}$$

Q6. Two-variable limits and continuity [4 marks]

Q6(a) [2 marks]

Find the limit as $(x,y) \to (0,0)$ or show that it does not exist by considering different paths of approach:

$$f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}.$$

Idea

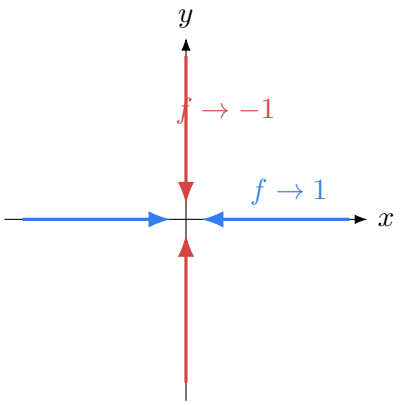
To prove a two-variable limit does *not* exist, it is enough to find two paths giving two different limiting values.

$$f(x,0) = \frac{x^2}{x^2} = 1 \qquad (x \neq 0),$$

so along the x-axis the limit is 1.

$$f(0,y) = \frac{-y^2}{y^2} = -1 \qquad (y \neq 0),$$

so along the y-axis the limit is -1.



Because the two paths give different values,

$$\boxed{\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2} \text{ does not exist.}}$$

Q6(b) [2 marks]

Define $f(0,0)$ so that

$$f(x,y) = \frac{\sin(3(x^2+y^2))}{x^2+y^2}$$

extends continuously to the origin. Justify your answer using polar coordinates.

Set

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Then $x^2 + y^2 = r^2$, so

$$f(x,y) = \frac{\sin(3r^2)}{r^2}.$$

Write

$$\frac{\sin(3r^2)}{r^2} = 3\frac{\sin(3r^2)}{3r^2}.$$

As $r \to 0$,

$$\frac{\sin(3r^2)}{3r^2} \to 1.$$

Therefore the limit is 3.

Define

$$\boxed{f(0,0) = 3}.$$

Then the value at the origin equals the limit, so the extended function is continuous there.

Q7. Differentials and engineering sensitivity [4 marks]

Q7

The tension T in the string of the yo-yo is

$$T(r, R) = \frac{mgr}{2r^2 + R^2},$$

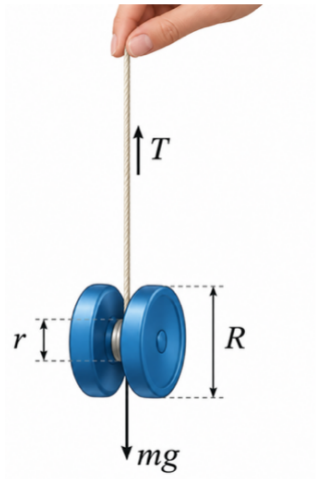
where m is the mass, g is the acceleration due to gravity, r is the inner radius and R is the outer radius. Assume m and g are constants.
The inner radius increases from 0.7 cm to 0.8 cm while the outer radius increases from 3.0 cm to 3.1 cm. Use differentials to estimate the change in tension.

a) Find $\partial T/\partial r$ and $\partial T/\partial R$.

b) Write dT in terms of dr and dR.

c) Determine dr and dR.

d) Given $(2r^2 + R^2)^2 = (9.98)^2 = 99.6004$ and $R^2 - 2r^2 = 8.02$ at $r = 0.7$ cm, $R = 3.0$ cm, evaluate dT. Does the tension increase or decrease?



Idea

For small changes in the inputs,

$$\Delta T \approx dT = T_r\, dr + T_R\, dR.$$

The signs of the two terms tell us how each radius affects the tension.

Using the quotient rule,

$$T_r = mg\frac{(2r^2 + R^2) - r(4r)}{(2r^2 + R^2)^2}$$
$$= mg\frac{R^2 - 2r^2}{(2r^2 + R^2)^2}.$$

Thus

$$\boxed{T_r = \frac{mg(R^2 - 2r^2)}{(2r^2 + R^2)^2}}.$$

Treating r as constant,

$$T_R = mgr\left[-(2r^2+R^2)^{-2}(2R)\right],$$

so

$$\boxed{T_R = -\frac{2mgrR}{(2r^2+R^2)^2}}.$$

$$\boxed{dT = \frac{mg(R^2-2r^2)}{(2r^2+R^2)^2}\,dr - \frac{2mgrR}{(2r^2+R^2)^2}\,dR.}$$

$$dr = 0.8 - 0.7 = 0.1 \text{ cm}, \qquad dR = 3.1 - 3.0 = 0.1 \text{ cm}.$$

Using the numerical values supplied in the question,

$$\begin{aligned} dT &= \frac{mg}{99.6004}\Big[(8.02)(0.1) - 2(0.7)(3.0)(0.1)\Big] \\ &= \frac{mg}{99.6004}(0.802 - 0.420) \\ &= \frac{0.382}{99.6004}\,mg \\ &\approx 0.003835\,mg. \end{aligned}$$

$$\boxed{dT \approx 3.84 \times 10^{-3}\,mg}$$

which is positive. Therefore the differential estimate predicts that the tension **increases**.

Part	Credit	Marks
(a)	Correctly derive $\partial T/\partial r$ and $\partial T/\partial R$ using the quotient rule	2
(b)	Correct total differential $dT = T_r\,dr + T_R\,dR$	$\frac{1}{2}$
(c)	Correctly identify $dr = 0.1$ cm and $dR = 0.1$ cm	$\frac{1}{2}$
(d)	Correct substitution/evaluation of dT and correct conclusion that tension increases	1

Partial credit indicated in the supplied rubric: starting the quotient rule may earn partial credit in (a); writing the total-differential formula before substitution earns partial credit in (b); each increment in (c) carries half of that part's credit; in (d), numerical evaluation and the sign/conclusion are separately creditable.

Do not confuse the finite change ΔT with dT. We are using dT as a first-order approximation: $\Delta T \approx dT$.

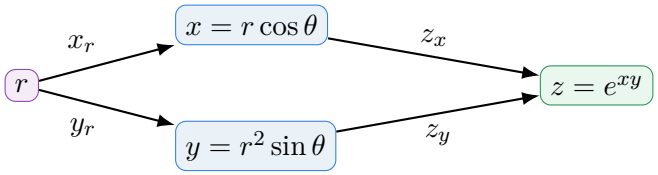
Q8. Multivariable Chain Rule [2 marks]

Q8

Let

$$z = e^{xy}, \qquad x = r\cos\theta, \qquad y = r^2\sin\theta.$$

Use the Chain Rule to obtain $\partial z/\partial r$. Also evaluate $\partial z/\partial r$ at $(r,\theta) = (1, \pi/2)$.



$$\boxed{z_r = z_x x_r + z_y y_r}.$$

Now

$$z_x = ye^{xy}, \qquad z_y = xe^{xy}, \qquad x_r = \cos\theta, \qquad y_r = 2r\sin\theta.$$

$$\begin{aligned} z_r &= ye^{xy}\cos\theta + xe^{xy}(2r\sin\theta) \\ &= e^{xy}(y\cos\theta + 2rx\sin\theta). \end{aligned}$$

Using $x = r\cos\theta$ and $y = r^2\sin\theta$,

$$xy = r^3\sin\theta\cos\theta,$$

and

$$y\cos\theta + 2rx\sin\theta = 3r^2\sin\theta\cos\theta.$$

Therefore

$$\boxed{z_r = 3r^2\sin\theta\cos\theta\, e^{r^3\sin\theta\cos\theta}}.$$

Since $\cos(\pi/2) = 0$,

$$z_r|_{(1,\pi/2)} = 0.$$

$$\boxed{\frac{\partial z}{\partial r} = 3r^2\sin\theta\cos\theta\, e^{r^3\sin\theta\cos\theta}}, \qquad \boxed{\left.\frac{\partial z}{\partial r}\right|_{(1,\pi/2)} = 0}.$$

Q9. Hessian matrix and equality of mixed partials [3 marks]

Q9

For

$$g(x,y) = e^{xy} + \sin(x+y) + x^2 y,$$

a) Find the Hessian matrix $H_g(x,y)$. Is it symmetric?

b) Is the Hessian matrix necessarily symmetric for any function $f(x,y)$? State the condition under which the mixed partial derivatives are guaranteed to be equal.

Idea

The Hessian is built from the four second-order partial derivatives:

$$H_g = \begin{pmatrix} g_{xx} & g_{xy} \\ g_{yx} & g_{yy} \end{pmatrix}.$$

$$g_x = ye^{xy} + \cos(x+y) + 2xy,$$
$$g_y = xe^{xy} + \cos(x+y) + x^2.$$

$$\boxed{g_{xx} = y^2 e^{xy} - \sin(x+y) + 2y},$$
$$\boxed{g_{xy} = (1+xy)e^{xy} - \sin(x+y) + 2x},$$
$$\boxed{g_{yx} = (1+xy)e^{xy} - \sin(x+y) + 2x},$$
$$\boxed{g_{yy} = x^2 e^{xy} - \sin(x+y)}.$$

$$\boxed{H_g(x,y) = \begin{pmatrix} y^2 e^{xy} - \sin(x+y) + 2y & (1+xy)e^{xy} - \sin(x+y) + 2x \\ (1+xy)e^{xy} - \sin(x+y) + 2x & x^2 e^{xy} - \sin(x+y) \end{pmatrix}}.$$

The off-diagonal entries are equal, so this Hessian is symmetric.

A Hessian is *not automatically* symmetric merely because the indicated second partials exist. A standard sufficient condition is that the mixed partial derivatives f_{xy} and f_{yx} are continuous in a neighbourhood of the point (equivalently, f is C^2 near the point). Then Clairaut's theorem gives

$$f_{xy} = f_{yx}.$$

For the given g, H_g is symmetric. In general, symmetry is guaranteed under the usual continuity condition on the mixed second partial derivatives.

Q10. Absolute extrema on a closed rectangle [4 marks]

Q10

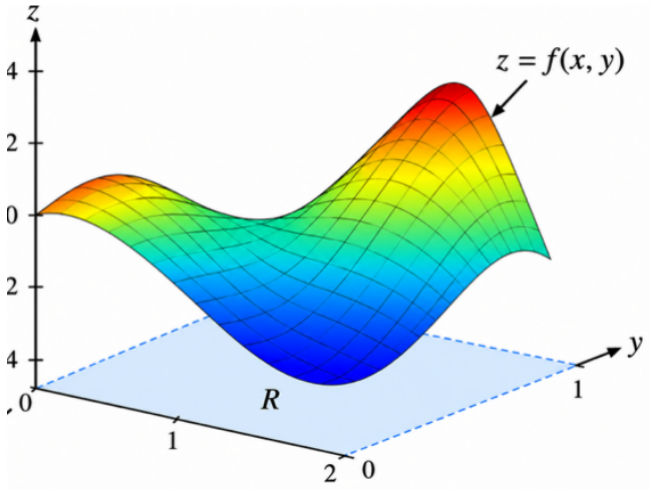
Find the absolute maximum and absolute minimum values of

$$f(x, y) = x^3 - 3xy + y^2$$

on

$$R = \{(x, y) \mid 0 \le x \le 2,\ 0 \le y \le 1\}.$$

Show all steps.



Idea

For a continuous function on a closed, bounded region, absolute extrema exist. To find them:

$$\boxed{\text{interior critical points} + \text{all boundary edges} + \text{corners.}}$$

$$f_x = 3x^2 - 3y = 3(x^2 - y), \qquad f_y = -3x + 2y.$$

Set both equal to zero:

$$y = x^2, \qquad y = \frac{3x}{2}.$$

Thus

$$x^2 = \frac{3x}{2} \quad \Longrightarrow \quad x = 0 \text{ or } x = \frac{3}{2}.$$

For $x = 3/2$, $y = 9/4 > 1$, so it is outside R. The point $x = 0, y = 0$ lies on the boundary. Hence there are no interior critical points in the open rectangle.

$$f(x, 0) = x^3, \qquad 0 \le x \le 2.$$

Hence

$$f(0, 0) = 0, \qquad f(2, 0) = 8.$$

$$f(x,1) = x^3 - 3x + 1.$$

Differentiate:

$$\frac{d}{dx}(x^3 - 3x + 1) = 3x^2 - 3 = 3(x^2 - 1).$$

The critical point in $[0,2]$ is $x = 1$. Therefore

$$f(0,1) = 1, \qquad f(1,1) = -1, \qquad f(2,1) = 3.$$

$$f(0,y) = y^2,$$

so on $0 \le y \le 1$ the values range from 0 to 1.

$$f(2,y) = 8 - 6y + y^2.$$

Its derivative is

$$2y - 6 < 0 \quad \text{for } 0 \le y \le 1,$$

so it decreases across the interval. Thus its endpoint values are

$$f(2,0) = 8, \qquad f(2,1) = 3.$$

Candidate	Point	Value
corner	$(0,0)$	0
corner	$(0,1)$	1
boundary critical	$(1,1)$	-1
corner	$(2,0)$	8
corner	$(2,1)$	3

$$\boxed{\text{Absolute minimum} = -1 \text{ at } (1,1)}$$

$$\boxed{\text{Absolute maximum} = 8 \text{ at } (2,0)}$$

For an absolute-extrema problem on a closed region, checking only $\nabla f = 0$ is incomplete. The boundary is essential; here both the minimum and maximum occur on the boundary.

Q11. Arc length of an astroid [3 marks]

Q11

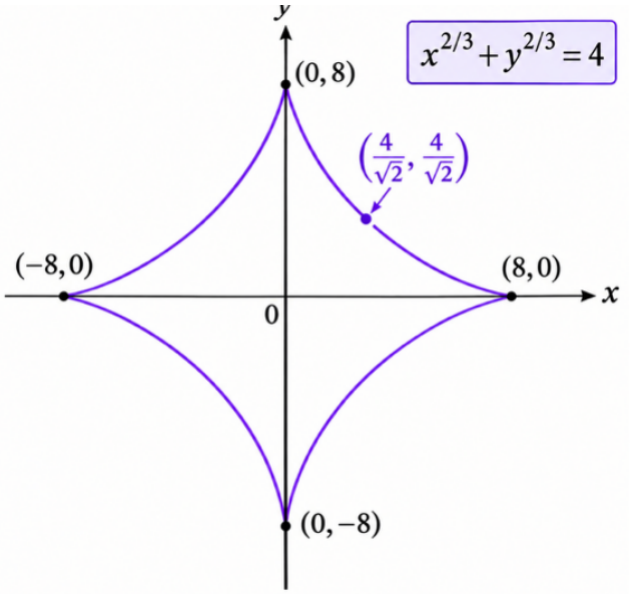
The curve is the astroid

$$x^{2/3} + y^{2/3} = 4.$$

Find the length of the arc joining

$$\left(\frac{4}{\sqrt{2}}, \frac{4}{\sqrt{2}}\right) \quad \text{and} \quad (8, 0).$$

This arc represents one-half of the first-quadrant portion, and one-eighth of the total length of the astroid.



Idea

Use the arc-length formula

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx.$$

The equation of the astroid lets the square root simplify beautifully.

From

$$x^{2/3} + y^{2/3} = 4,$$

differentiate with respect to x:

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y' = 0.$$

Hence

$$y' = -\frac{y^{1/3}}{x^{1/3}} = -\left(\frac{y}{x}\right)^{1/3}.$$

$$\begin{aligned} 1 + (y')^2 &= 1 + \frac{y^{2/3}}{x^{2/3}} \\ &= \frac{x^{2/3} + y^{2/3}}{x^{2/3}} \\ &= \frac{4}{x^{2/3}}. \end{aligned}$$

Because $x > 0$ on the arc,

$$\sqrt{1 + (y')^2} = \frac{2}{x^{1/3}}.$$

The two endpoints have

$$x = \frac{4}{\sqrt{2}} = 2\sqrt{2} \quad \text{and} \quad x = 8.$$

Therefore

$$L = \int_{2\sqrt{2}}^{8} 2x^{-1/3}\, dx.$$

$$\begin{aligned} L &= 2\left[\frac{x^{2/3}}{2/3}\right]_{2\sqrt{2}}^{8} \\ &= 3\left[x^{2/3}\right]_{2\sqrt{2}}^{8}. \end{aligned}$$

Now

$$8^{2/3} = 4,$$

and since $2\sqrt{2} = 2^{3/2}$,

$$(2\sqrt{2})^{2/3} = 2.$$

Thus

$$L = 3(4 - 2) = 6.$$

$$\boxed{L = 6}$$

The question also tells us that this is one-eighth of the total astroid length, so as a check the full perimeter would be $8 \times 6 = 48$.

Credit	Marks
Write $y = (4 - x^{2/3})^{3/2}$ and obtain the correct dy/dx	1
Obtain $\sqrt{1 + (dy/dx)^2} = 2x^{-1/3}$	$\frac{1}{2}$
Set up the arc-length integral with limits $2\sqrt{2}$ to 8	$\frac{1}{2}$
Integrate to $3x^{2/3}$	$\frac{1}{2}$
Substitute the limits and obtain $L = 6$	$\frac{1}{2}$

One-look answer summary

Q	Final result
1(a)	No; the limit can exist even if f is undefined at the point.
1(b)	No; existence of f_x, f_y alone does not imply differentiability.
2	Domain $x^2+y^2>36$; boundary $x^2+y^2=36$; domain open; range $(0,\infty)$.
3	Highest peak A; steepest slope at C.
4	$z_x(1,1,1)=1$.
5	$L(x,y,z)=x-y+z$.
6(a)	Limit does not exist (1 along $y=0$, -1 along $x=0$).
6(b)	Define $f(0,0)=3$.
7	$T_r=\dfrac{mg(R^2-2r^2)}{(2r^2+R^2)^2}$, $T_R=-\dfrac{2mgrR}{(2r^2+R^2)^2}$, $dT\approx 0.003835\,mg>0$; tension increases.
8	$z_r=3r^2\sin\theta\cos\theta\, e^{r^3\sin\theta\cos\theta}$; at $(1,\pi/2)$ the value is 0.
9	Hessian shown above; symmetric for this g; mixed partials are guaranteed equal when they are continuous in a neighbourhood.
10	Absolute minimum -1 at $(1,1)$; absolute maximum 8 at $(2,0)$.
11	Arc length $\boxed{6}$.

For this paper, the recurring habits are: translate words into mathematical conditions, show one clean reason for every conclusion, and when a multivariable problem involves a closed region or several possible paths, do not check only the most obvious case.